

Artificial Intelligence & Machine Learning



Course Curriculum

Artificial Intelligence

Module 1: Introduction to Data Science

- What is Data Science?
- What is Machine Learning?
- What is Deep Learning?
- What is AI?
- Data Analytics & it's types

Module 2: Introduction to Python

- What is Python?
- Why Python?
- Installing Python
- Python IDEs
- Jupyter Notebook Overview

Module 3: Python Basics

- Python Basic Data types
- Lists
- Slicing
- IF statements
- Loops

- Dictionaries
- Tuples
- Functions
- Array
- Selection by position & Labels

Module 4: Python Packages

- Pandas
- Numpy
- Sci-kit Learn
- Mat-plot library

Module 5: Importing Data

- Reading CSV files
- Saving in Python dat
- Loading Python data objects
- Writing data to csv file

Module 6: Manipulating Data

- Selecting rows/observations
- Rounding Number
- Selecting columns/fields
- Merging data

- Data aggregation
- Data munging techniques

Module 7: Statistics Basics

- **Central Tendency**
 - Mean
 - Median
 - Mode
 - Skewness
 - Normal Distribution
- **Probability Basics**
 - What does mean by probability?
 - Types of Probability
 - Mode
- **Standard Deviation**
 - Data deviation & distribution
 - Variance
- **Bias variance Trade off**
 - Underfittin
 - Overfitting
- **Distance metrics**
 - Euclidean Distance
 - Manhattan Distance

- **Outlier analysis**
 - What is an Outlier?
 - Inter Quartile
 - Range
 - Box & whisker plot
 - Upper Whisker
 - Lower Whisker
 - Scatter plot
 - Cook's Distance
- **Missing Value treatment**
 - What is a NA?
 - Central Imputation
 - KNN imputation
 - Dummification
- **Correlation**
 - Pearson correlation
 - ositive & Negative correlation

Module 8: Error Metrics

- **Classification**
 - Confusion Matrix
 - Precision
 - Recall

- Dummification
- Specificity
- F1 Score
- **Regression**
 - MSE
 - RMSE
 - MAPE

Module 9: Machine Learning

- **Supervised Learning**
 - **Linear Regression**
 - Linear Equation
 - Slope
 - Intercept
 - R square value
 - **Logistic regression**
 - ODDS ratio
 - Probability of success
 - Probability of failure Bias Variance Tradeoff
 - ROC curve
 - Bias Variance Tradeoff curve
- **Unsupervised Learning**

- K-Means
- K-Means ++
- Hierarchical Clustering
- SVM
- Support Vectors
- Hyperplanes
- 2-D Case
- Linear Hyperplane
- SVM Kernal
- Linear
 - polynomial
- Other Machine Learning algorithms
- K - Nearest Neighbour
- Naïve Bayes Classifier
- Decision Tree — CART
- Decision Tree - C50
- Random Forest

Module 10: Artificial Intelligence Introduction

- Perceptron
- Multi-Layer perceptron
- Markov Decision Process
- Logical Agent & First Order Logic

- AL Applications

Module 11: Deep Learning Algorithms

- CNN - Convolutional Neural Network
- RNN - Recurrent Neural Network
- ANN - Artificial Neural Network
- Introduction to NLP
- Text Pre-processing
 - Noise Removal
- Lexicon Normalization
 - Lemmatization
- Stemming
 - Object Standardization
- Text to Features (Feature Engineering)
- Syntactical Parsing
 - Dependency Grammar
- Part of Speech Tagging
 - Entity Parsing
- Named Entity Recognition
 - Topic Modelling
- N-Grams
- TF-IDF
- Frequency / Density Features

- Word Embedding's o Tasks of NLP
- Text Classification o Text Matching
- Levenshtein Distance o Phonetic Matching
- Flexible String Matching

Course Duration : 2 Months